SIYUAN HE

he662@purdue.edu | Ph.D. candidate in Computer Science (looking for intern and jobs) | sweetsinpackets.github.io

EDUCATION

- **Purdue University, West Lafayette, IN**, Ph.D. candidate in Computer Science *Aug. 2021 - TBD*
- **University of Michigan, Ann Arbor, MI**, Master of Science in Information *Aug. 2019 - Apr. 2021*
- **Shanghai Jiaotong University, Shanghai**, Bachelor of Electr. and Comput. Eng. *Sept. 2016 - Aug. 2020*

SELECTED RESEARCH PROJECTS

Reachability Types
Advised by Prof. Tiark Rompf, in collaboration with group members, Related tools: Coq/Rocq *Sept. 2021 - present*
- We design resource tracking type systems for higher-order functional programs with System F<: style polymorphism and first-class references, statically reasoning about separation and controlled mutable sharing, ensuring no data racing. [OOPSLA 25]
- We adopt first-class shadow arenas to allow coarse-grained resource tracking and multi-hop store cycles, selective lexical or non-lexical lifetime, and flow-insensitive deallocation reasoning. [OOPSLA 26, first author]
- We extend declarative typing with algorithmic type-checking and inference [PLDI 26].

Typestate via Revocable Capabilities
Advised by Prof. Tiark Rompf, Contributed as co-author *May. 2025 - Oct. 2025*
- We propose a programming pattern with flow-sensitive typestate, decoupling typestate from lexical scopes, allowing functions to receive, revoke and return capabilities. Implemented in Scala 3 compiler, using path-dependent types and implicits. [PLDI 26]

Boundless Quantification for Lightweight Polymorphism via Projection Types
Advised by Prof. Tiark Rompf, Related tools: Coq/Rocq *Sept. 2024 - present*
- We study boundless quantification for arrow-shaped formal parameters in System F<: via projection types, allowing lightweight polymorphism for arrow types, latent effects (potentially) without parametric polymorphism. [in submission, first author]
- We develop logical relations for higher-kind type systems with subtyping (Fomega<:) with delayed interpretation. [artifact online]
- We encode lightweight dependent types via image types and effect polymorphism without exposing latent effects. [in progress]

Binary Auditing with LLM
Contributor *Dec. 2025 - Mar. 2026*
- We adopt LLM to verify neural-symbolic auditing of firmware binaries for exploitable vulnerabilities. [in submission]

Formal verification of a pathway finding algorithm
Advised by Prof. Jean-Baptiste Jeannin, Related tools: Coq/Rocq *Oct. 2019 - Oct. 2020*
- We formalize an airport taxiway path-finding algorithm and formally verify its correctness. [DASC 20, first author]

WORK EXPERIENCE

Arista Networks, CVP Team
Software Engineer Intern. Related tools: Golang *May 2024 - Aug. 2024*
- Developed a SHA512 Digest validation for cloud image downloading.
- Developed a version filter for old images without guaranteed digest support.
- Designed the interaction between front-end and back-end to perform error highlighting and back-track navigation.

Nokia Shanghai-Bell - ION Department, 5G Core Team
Software Engineer Intern. Related tools: C++ *May 2020 - Aug. 2020*
- Adapted a test framework and devised comprehensive test cases to simulate extensive user-to-server random access scenarios.
- Executed laboratory assessments on an experimental network server hardware and optimized Identity Verification Algorithm.

Lenovo Research - AI Lab, Face Recognition Team
Research Intern. Related tools: Python, C++ *Dec. 2017 - Feb. 2018*
- Developed image pre-processing modules (blur, over-exposure, similarity) and integrated in a deployed face recognition system.

SERVICES AND TALKS

- Artifact Evaluation Committee: OOPSLA 25
- Invited Talk at IWACO 26 (upcoming)

TEACHING AND COURSES

- Operating Systems. CS 352 (2022 Fall, 2023 Spring, 2023 Fall), CSCI 403 (2026 Spring Indy Campus).
- Architecture. CSCI 402 (2025 Fall Indy Campus). VE 370 (2019 Summer, Shanghai Jiaotong University).
- C Programming. CS 159 (2024 Fall, 2025 Spring). CS 240 (2025 Summer). *Graduate Teaching Award*
- Compilers. CS 352 (2024 Spring).

SELECTED PROJECTS

- **Discord Bot for Raid Management** (2021): a Python bot managing online-game raid battles with member registration, battle-log archival, and real-time status reporting.